

Name: Potestades, North Nygel G.	Date Performed: 11/14/25
Course/Section: CPE31S2	Date Submitted: 11/20/25
Instructor: Engr. Robin Valenzuela	Semester and SY: 1st Semester 2025-2026

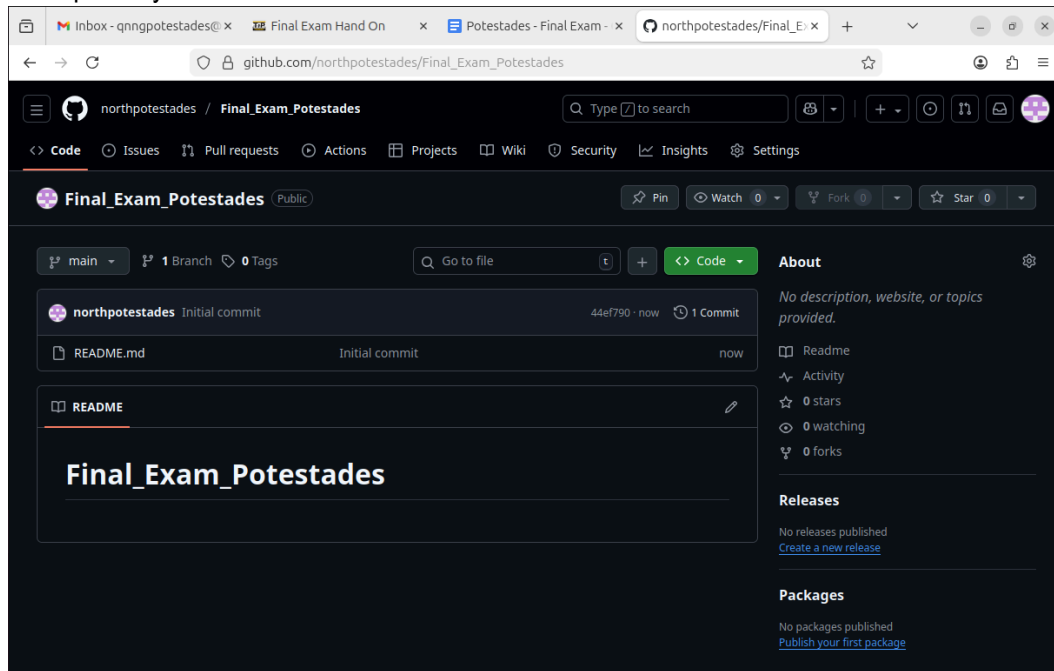
Final Exam Hand On

I. Tools Needed:

1. VM with Ubuntu, CentOS and Ansible installed
2. Web browser

II. Procedure:

1. Create a repository and label it as "Final_Exam_Surname"



Here we can see the new repository created for this final hands-on skills exam. The link to the repository is pasted in question 6.

2. Clone your new repository in your VM

```

north@workstation:~$ git clone git@github.com:northpotestades/Final_Exam_Potestades.git
Cloning into 'Final_Exam_Potestades'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
north@workstation:~$ ls
CPE212_H0A11_Potestades  Documents          Music              snap
CPE212_Potestades        Downloads          Pictures           Templates
Desktop                  Final_Exam_Potestades  Public            Videos
north@workstation:~$ cd Final_Exam_Potestades
north@workstation:~/Final_Exam_Potestades$ ls
README.md

```

Here we can see the repository cloned in my VM, allowing me to push files into the repository.

3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers (Chosen = Apache2)

3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host) (Chosen = Nagios)

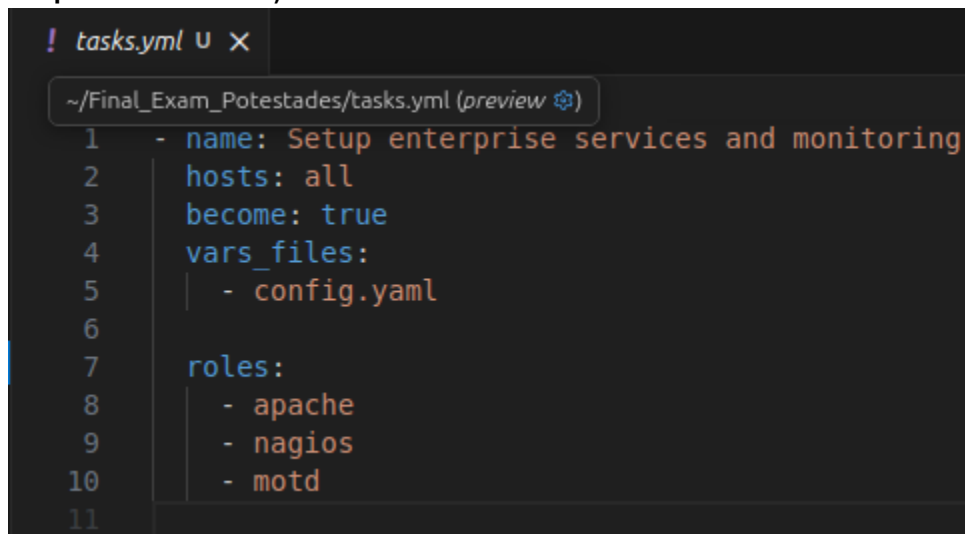
3.3 Change Motd as "Ansible Managed by <username>"

4. Push and commit your files in GitHub

```
• north@workstation:~/Final_Exam_Potestades$ git add --all
• north@workstation:~/Final_Exam_Potestades$ git commit -m "Final Update Final Exam"
[main 43bcd14] Final Update Final Exam
 9 files changed, 149 insertions(+)
 create mode 100644 roles/apache/tasks/main.yml
 create mode 100644 roles/apache/templates/index.html.j2
 create mode 100644 roles/motd/tasks/main.yml
 create mode 100644 roles/nagios/handlers/main.yml
 create mode 100644 roles/nagios/tasks/main.yml
 create mode 100644 roles/nagios/templates/nagios.cfg.j2
 create mode 100644 tasks.yml
• north@workstation:~/Final_Exam_Potestades$ git push origin main
Enumerating objects: 24, done.
Counting objects: 100% (24/24), done.
Delta compression using up to 4 threads
Compressing objects: 100% (14/14), done.
Writing objects: 100% (21/21), 2.69 KiB | 550.00 KiB/s, done.
Total 21 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:northpotestades/Final_Exam_Potestades.git
 b05b592..43bcd14  main -> main
```

5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)

Input (tasks separated into roles)



```
! tasks.yml U X
~/Final_Exam_Potestades/tasks.yml (preview ⚙)
1  - name: Setup enterprise services and monitoring
2    hosts: all
3    become: true
4    vars_files:
5      - config.yaml
6
7    roles:
8      - apache
9      - nagios
10     - motd
11
```

Process

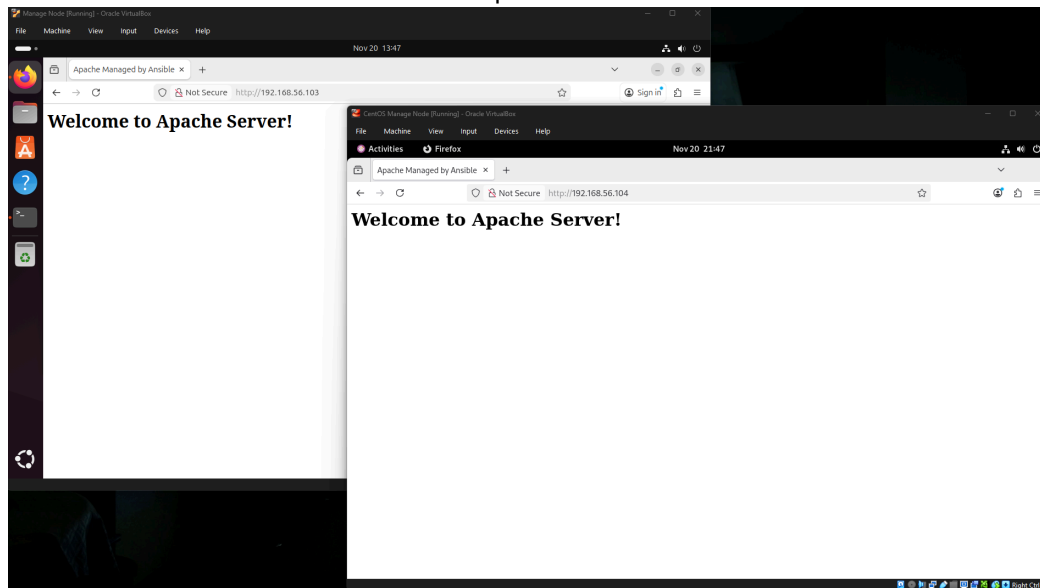
```
north@workstation:~/Final_Exam_Potestades$ ansible-playbook tasks.yml -e @config.yaml --ask-become-pass

RUNNING HANDLER [nagios : Restart Apache] *****
changed: [192.168.56.103]

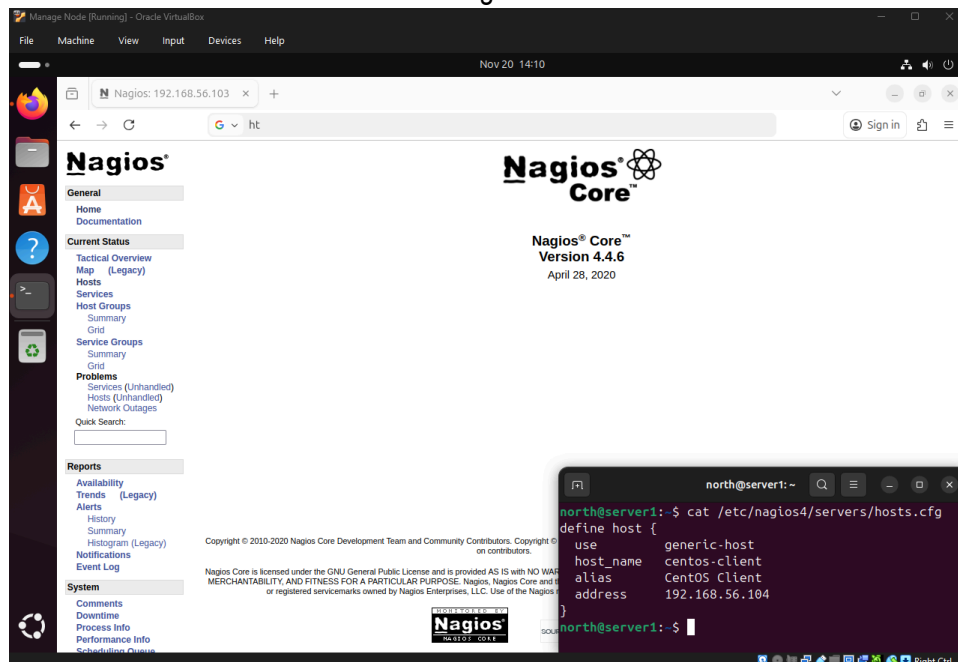
PLAY RECAP *****
192.168.56.103      : ok=16  changed=6  unreachable=0  failed=0  skipped=2
   rescued=0  ignored=0
192.168.56.104      : ok=7   changed=0  unreachable=0  failed=0  skipped=10
   rescued=0  ignored=0
```

Output

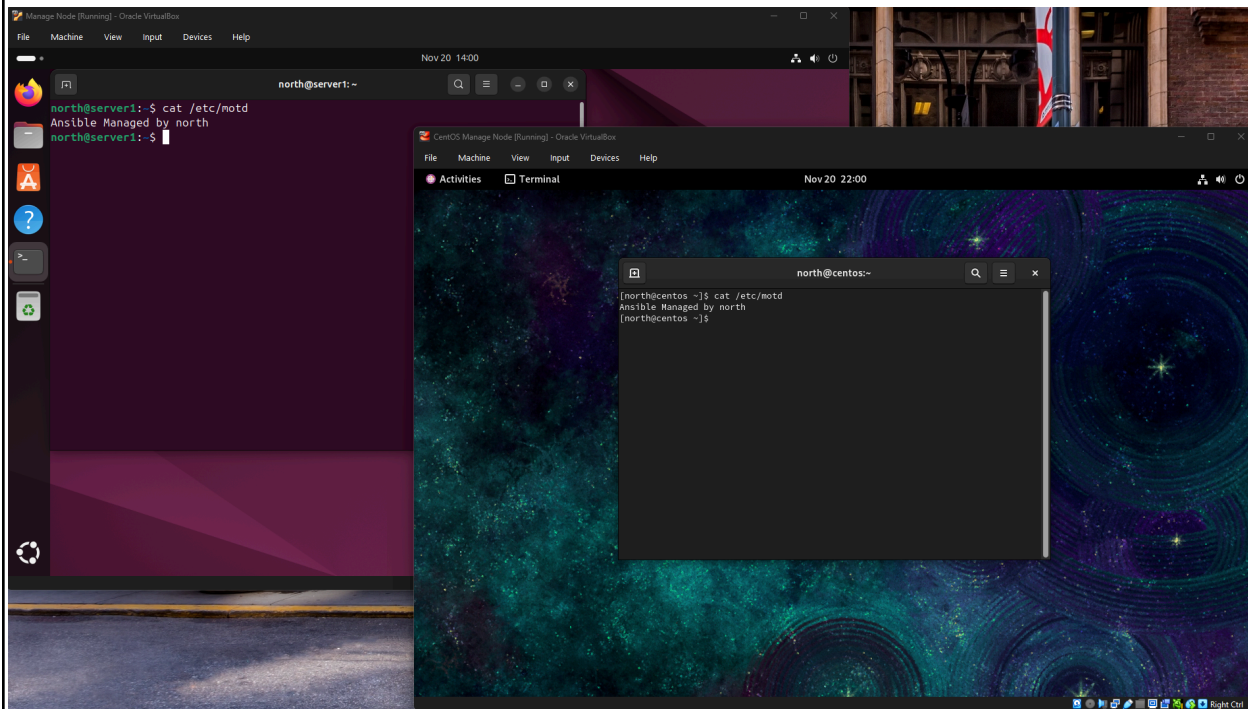
Apache2



Nagios



MOTD



6. For your final exam to be counted, please paste your repository link as an answer in this exam.

https://github.com/northpotestades/Final_Exam_Potestades

Note: Extra points if you will implement the said services via containerization.